

A sample sum

A sample sum totals the observations in a sample.

$$S_n = X_1 + \cdots + X_n$$

For shipment weight, would we need an individual mass, sample mean, or sample sum?

Mean and sum

A sample sum equals the sample size times its mean.

$$S_n = n\bar{X}$$

If 40 packages have mean mass 500 g, what is their total mass?

The center of sums

For n observations from a population with mean μ , the expected sum is $n\mu$.

$$E(S_n) = n\mu$$

If one item has mean mass 50 g, what is the expected mass of 100 items?

The spread of sums

For n independent observations with common SD σ , the sum has SD $\sigma\sqrt{n}$.

$$\text{SD}(S_n) = \sigma\sqrt{n}$$

If $\sigma=4$ g and $n=100$, what is the SD of total mass?

Recovering sample size

Synthetic independent observations have $\sigma = 6$. Their sum has standard deviation 42.

What sample size produced this sum?

Independent variation

Model: individual item masses have SD 4 g;
100 items are independent.

Why would using 100×4 g as the SD of total mass overstate its spread?

Increasing the total sample

Model: independent items have mean mass 50 g and SD 4 g.

How do the mean and SD of the sum change when n rises from 100 to 400?

A sum can be normal

A sum of independent normal observations is normal for every sample size.

What model describes the total of 100 independent $N(50,4)$ masses?

A normal approximation for sums

The central limit theorem for sums makes totals of many independent, identically distributed observations approximately normally distributed when variance is finite.

What information about strong skewness would you want before using that approximation?

Standardizing a total

The z-score of a sum uses the sum's mean and standard deviation.

$$z = \frac{\sum x - n\mu}{\sigma\sqrt{n}}$$

For $\mu=50$, $\sigma=4$, and $n=100$, what z-score belongs to a total of 5,080 g?

A total above a limit

Model: 100 independent item masses follow $N(50,4)$, in grams.

What is the probability that their total exceeds 5,080 g?

A total below a limit

Model: 100 independent item masses follow $N(50,4)$, in grams.

What is the probability that their total is below 4,960 g?

A total within a range

Model: 100 independent item masses follow $N(50,4)$, in grams.

What is the probability that their total lies between 4,940 and 5,060 g?

A capacity cutoff

A percentile of sums is a cutoff for total values. Model: 100 independent masses follow $N(50,4)$, in grams.

What capacity would exceed the total mass in 99% of shipments?

A specified distance for a sum

Model: independent observations have mean 50 g and SD 4 g. Each sample contains 100 observations.

What total lies 1.5 standard deviations above the mean of the sums?

An equivalent event

A shipment contains exactly 100 items.

What event involving the sample mean is equivalent to a total above 5,080 g?

The same standardized distance

Model: 100 independent masses follow $N(50,4)$. A total is 5,080 g.

Do the sum and its corresponding mean have different z-scores?

Sums versus averages

Model: individual masses have SD 4 g.
Compare independent samples of 25 and 100.

Does increasing n make sums or averages less variable in their original units?

A skewed population total

Model: independent task durations are exponential with mean and SD 6 minutes. A batch has 100 tasks.

What approximate normal model describes total duration?

Checking dependence

A shared machine adjustment can shift every item's mass in the same shipment.

Would the independent-sum SD formula still be justified?

Defining the correct total

Model: item masses average 50 g. Each shipment adds a fixed 200 g container.

What is the expected total mass of a container holding 100 items?